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PlanarCityLoc: Breaking Silhouette Ambiguity in LoD Aerial Localization via 3D Planes

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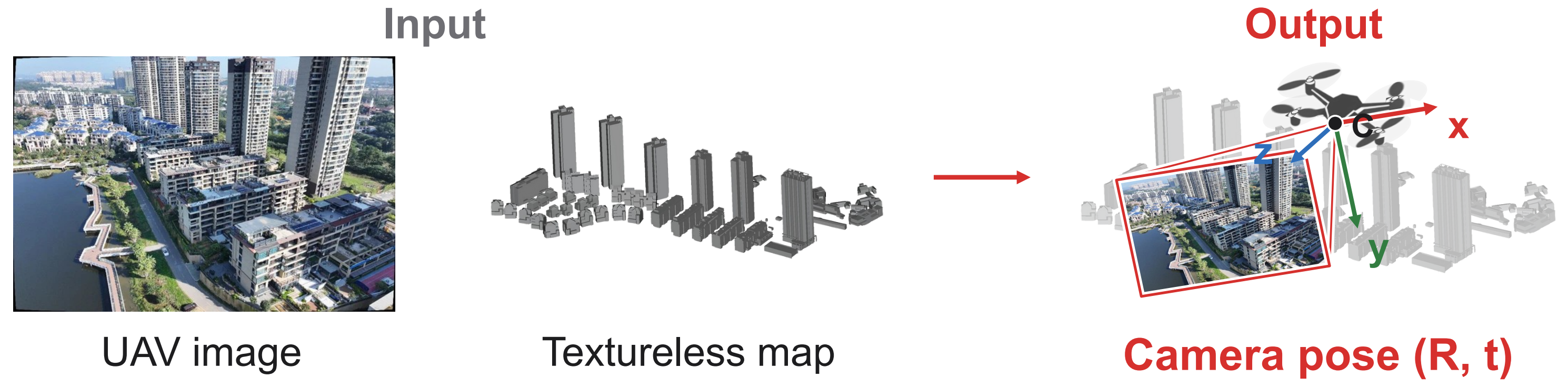
TL;DR

We present PlanarCityLoc to locate a drone in a 3D city map: building outlines alone can fit different viewpoints. 3D planes help resolve this silhouette ambiguity; our results improve recall over off-the-shelf outline matching.

Goal & motivation

Goal: 6-DoF UAV localization in LoD maps



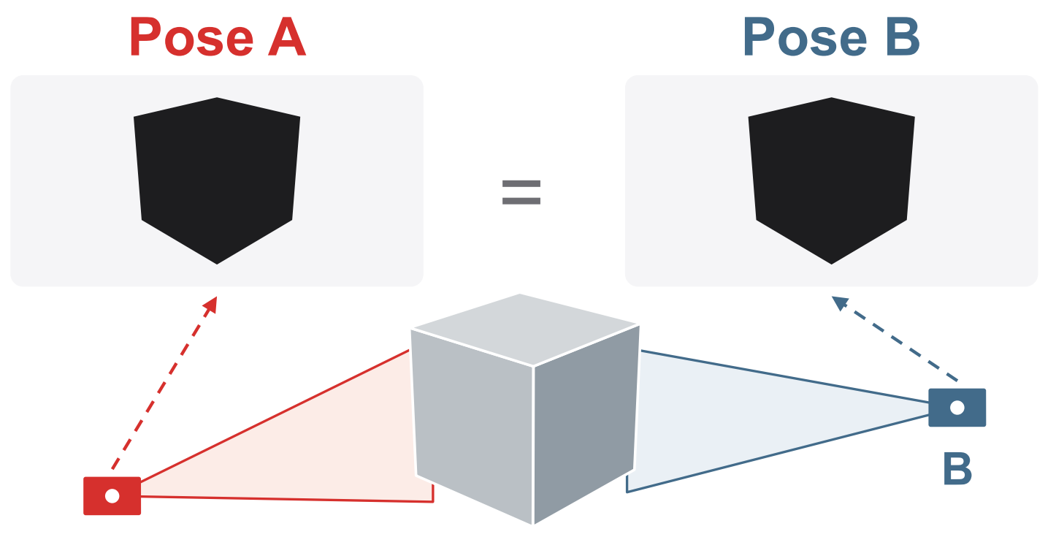
+ intrinsics and a coarse GPS/IMU prior

Why? Reliable positioning for UAV applications

Urban mapping **Building inspection** **Urban navigation**

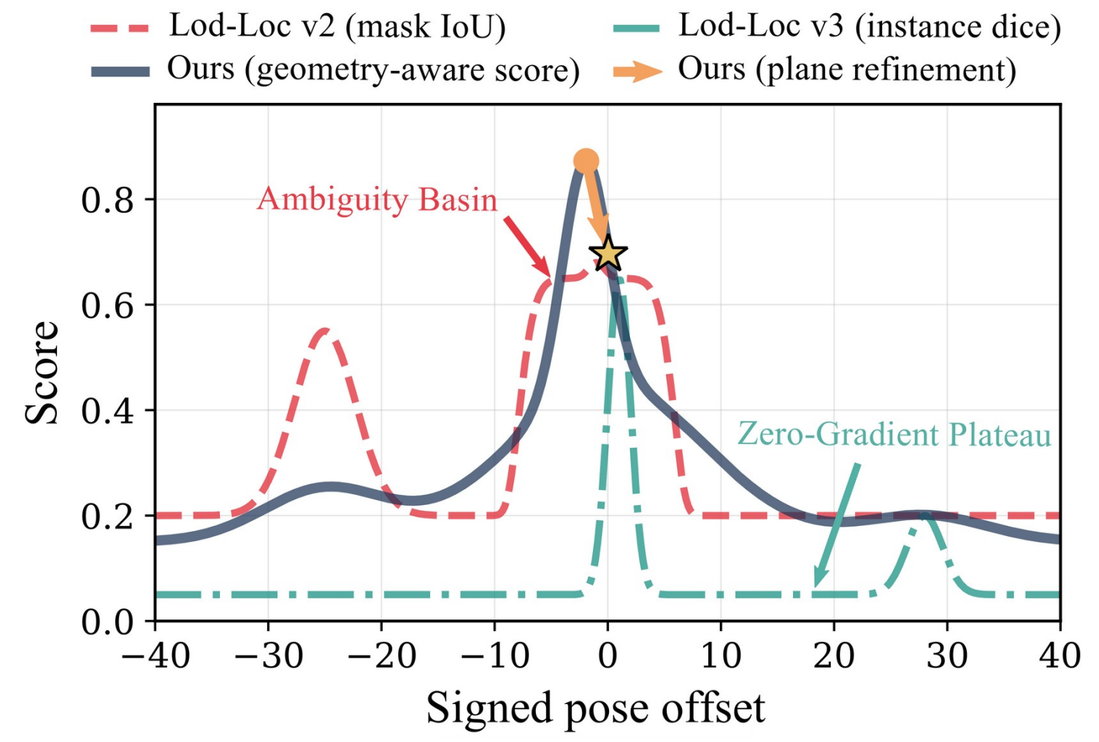
Application illustrations

Limitation: silhouette ambiguity in LoD-Loc v2 [3] and LoD-Loc v3 [4]



Symmetric block · schematic

Different poses can share the same mask.
Mask overlap supplies no 3D matches.
Disjoint masks also give flat scores.



Illustrative optimization landscape

Contributions & takeaway

① **Planar grounding**
3D map patches ↔ 2D image regions.

② **Geometry-aware pose search**
Masks, edges and depth guide initialization.

③ **Off-the-shelf visual models**
No dataset-specific model fine-tuning.

Address silhouette ambiguity through 3D plane grounding.

Method

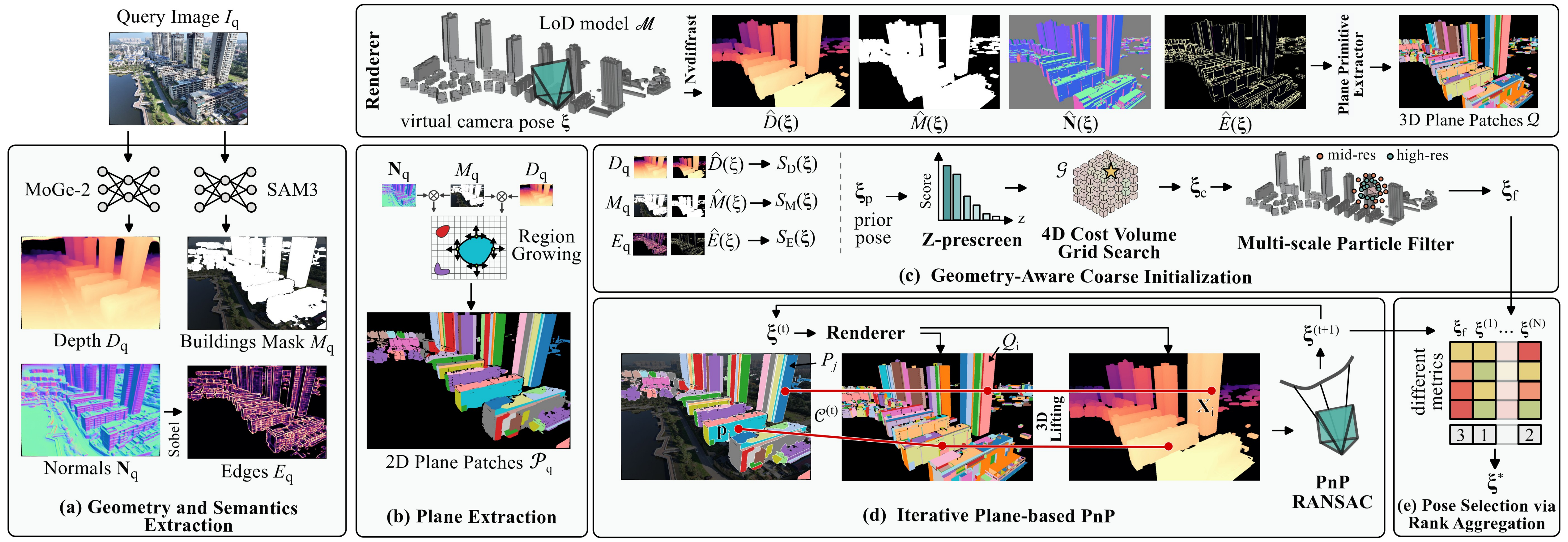
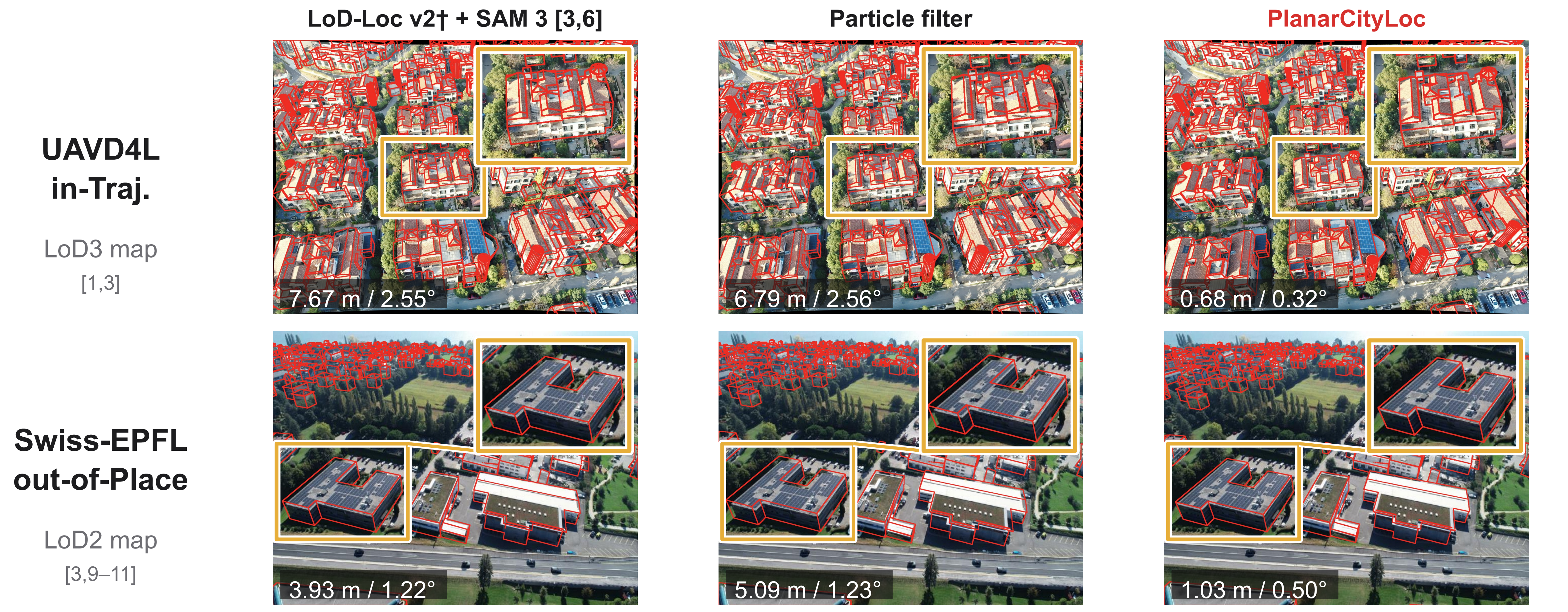

Query Image I_q
Renderer
LoD model $\mathcal{M}$
virtual camera pose ξ
Nvdiffrast
$\hat{D}(\xi)$
$\hat{M}(\xi)$
$\hat{N}(\xi)$
$\hat{E}(\xi)$
Plane Primitive Extractor
3D Plane Patches $\mathcal{Q}$
MoGe-2
SAM3
Depth D_q
Buildings Mask M_q
Normals N_q
Sobel
Edges E_q
(a) Geometry and Semantics Extraction
N_q
M_q
D_q
Region Growing
2D Plane Patches $\mathcal{P}_q$
(b) Plane Extraction
D_q $\hat{D}(\xi) \rightarrow S_D(\xi)$
M_q $\hat{M}(\xi) \rightarrow S_M(\xi)$
E_q $\hat{E}(\xi) \rightarrow S_E(\xi)$
ξ_p prior pose
Score
z
Z-prescreen
$\mathcal{G}$
4D Cost Volume Grid Search
ξ_c
mid-res
high-res
Multi-scale Particle Filter
ξ_f
(c) Geometry-Aware Coarse Initialization
$\xi^{(t)}$ → Renderer
P_j
$\mathbf{p}_j$
$\mathcal{C}^{(t)}$
Q_i
3D Lifting
$\mathbf{X}_i$
PnP RANSAC
$\xi^{(t+1)}$
(d) Iterative Plane-based PnP
ξ_f $\xi^{(1)}$… $\xi^{(N)}$
different metrics
3
1
2
ξ^*
(e) Pose Selection via Rank Aggregation

Image cues: SAM 3 [6] and MoGe-2 ViT-S [7]. Robust refinement: plane-based PnP–RANSAC [13].

Qualitative localization



Red: projected LoD edges. Labels: translation / rotation error.

Localization recall

Joint recall at 2 m / 2° (%) ↑

	UAVD4L-LoDv2 [1,3]		Swiss-EPFL-v2 [3,9–11]	
Method	**in-Traj.**	**out-Traj.**	**in-Place**	**out-Place**
Trainable methods				
LoD-Loc · trained [2]	49.6	54.2	36.8	14.2
LoD-Loc v2† · oracle GT [3]	**57.2**	**64.9**	**74.9**	**43.3**
LoD-Loc v3† · oracle GT [4]	54.5	41.3	56.3	40.1
Off-the-shelf methods				
MC-Loc · DINOv2 [5,12]	1.2	2.4	0.9	2.9
LoD-Loc v2† · SAM 3 [3,6]	35.7	64.3	9.4	17.4
Cost volume (ours)	27.6	34.9	15.4	11.3
+ particle filter (ours)	36.0	60.8	**24.9**	14.5
PlanarCityLoc	**75.4**	**80.6**	20.6	**48.5**

† Adapted re-runs: GT = oracle masks (not trained predictions); SAM 3 = off-the-shelf predictions.
LoD-Loc [2]: published trained result. Map/search setups differ. Published trained v2/v3 results: see paper.
Bold/underline: best/second within oracle or off-the-shelf runs, not across setups.

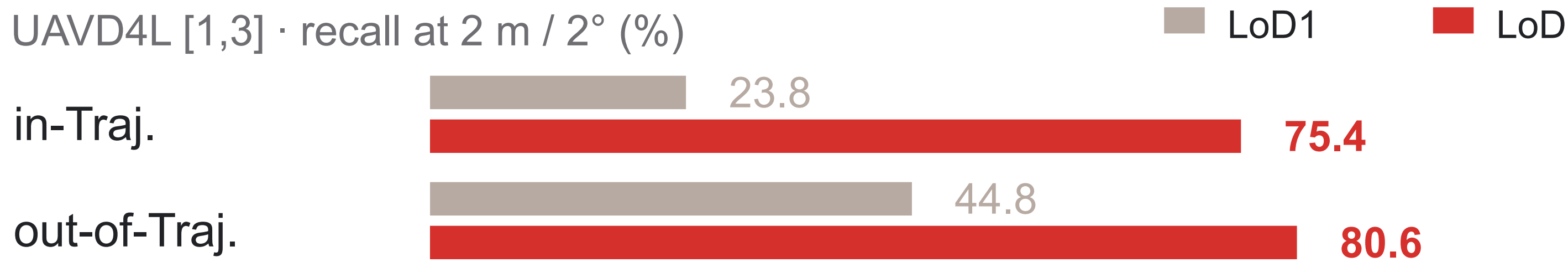
Pose accuracy

Split	3 m / 3° ↑	5 m / 5° ↑	T.e. (m) ↓	R.e. (°) ↓
UAV · in-Traj.	90.8	97.2	1.28	0.68
UAV · out-Traj.	95.1	99.0	1.26	0.69
Swiss · in-Place	39.0	59.7	3.70	1.90
Swiss · out-Place	72.3	89.4	1.98	1.21

PlanarCityLoc: recall (%) and median translation / rotation errors.

Ablations

Map detail improves pose accuracy



Depth and normals matter

Depth + normal predictor	in-Traj.	out-of-Traj.
Metric3D v2 · ViT-S [8]	33.0	55.0
MoGe-2 · ViT-S [7]	**75.0**	**84.0**

Geometry: 100 queries/split, SAM 3 [6] fixed. Map detail: full splits.

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Application illustrations: AI-generated. All scientific figures and results use the paper's recorded sources.